

THE SECRET UNDERSEA

Inside the Advanced Technology Vessel, India's hush-hush project to build an atomic-powered ballistic

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Located up the winding shipping channel in Visakhapatnam harbour is a secret, completely enclosed facility known only as the Shipbuilding Centre (SBC). Inside this dry dock, nearly 50m below ground level, is a cylindrical black shape, which is as tall as a two-storey building and at 104 m in length, is longer than the Qutub Minar lying on its side. Technicians working on it confess to a surge of national pride: India's first nuclear-powered ballistic missile submarine or SSBN is arguably its greatest engineering project.

For over a quarter of a century, the Advanced Technology Vessel (ATV), smaller than the *USS Alabama* from *Crimson Tide*, has been among the

most highly-classified government programmes, if not the most delayed. Officials still refuse to confirm the existence of the project or the sea-based ballistic missile. A decade after India came out of the nuclear closet in the sands of Pokhran, it has moved some tantalising steps closer to realising the third and possibly the toughest of the three legs of the triad enunciated in its nuclear doctrine: a sea-based deterrent or a secure underwater platform for launching nuclear weapons. "Things are developing as per schedule," Defence Minister A.K. Antony recently said of ATV. Early last month, Chief of the Naval Staff Admiral Sureesh Mehta was the first government official to not only confirm its existence but also lay down a timeframe: "It is a DRDO project and a tech-

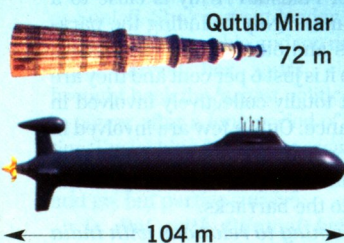
nology demonstrator. It is somewhere near completion and will be in the water in two years."

The admiral had reason to feel confident about the project. Just last month, an 80MW nuclear reactor, smaller than a bus, was pushed into the hull of the submarine and successfully integrated—a milestone in the project approved by the then prime minister Indira Gandhi in 1970. By April 2009, the submarine will be launched and will begin sea trials before it is inducted into the navy. The goal is to field a fleet of three SSBNs by 2015, one in reserve and two on patrol, each carrying 12 nuclear-tipped ballistic missiles. Possibly the last "gift" to India from the now-extinct Soviet Union, it was designed with Russian assistance in the late '80s. Based on an entirely new design, the

BABY BOOMER

India's nuclear-propelled ballistic missile submarine (artist's impression below) will be the world's smallest but it will pack a mega-tonne punch

DIMENSIONS



BUILDING AN SSBN COMBINES THE MAXIMUM NUMBER OF DISCIPLINES

-  Marine engineering
-  Nuclear engineering
-  Missile technology
-  Nuclear weaponry

